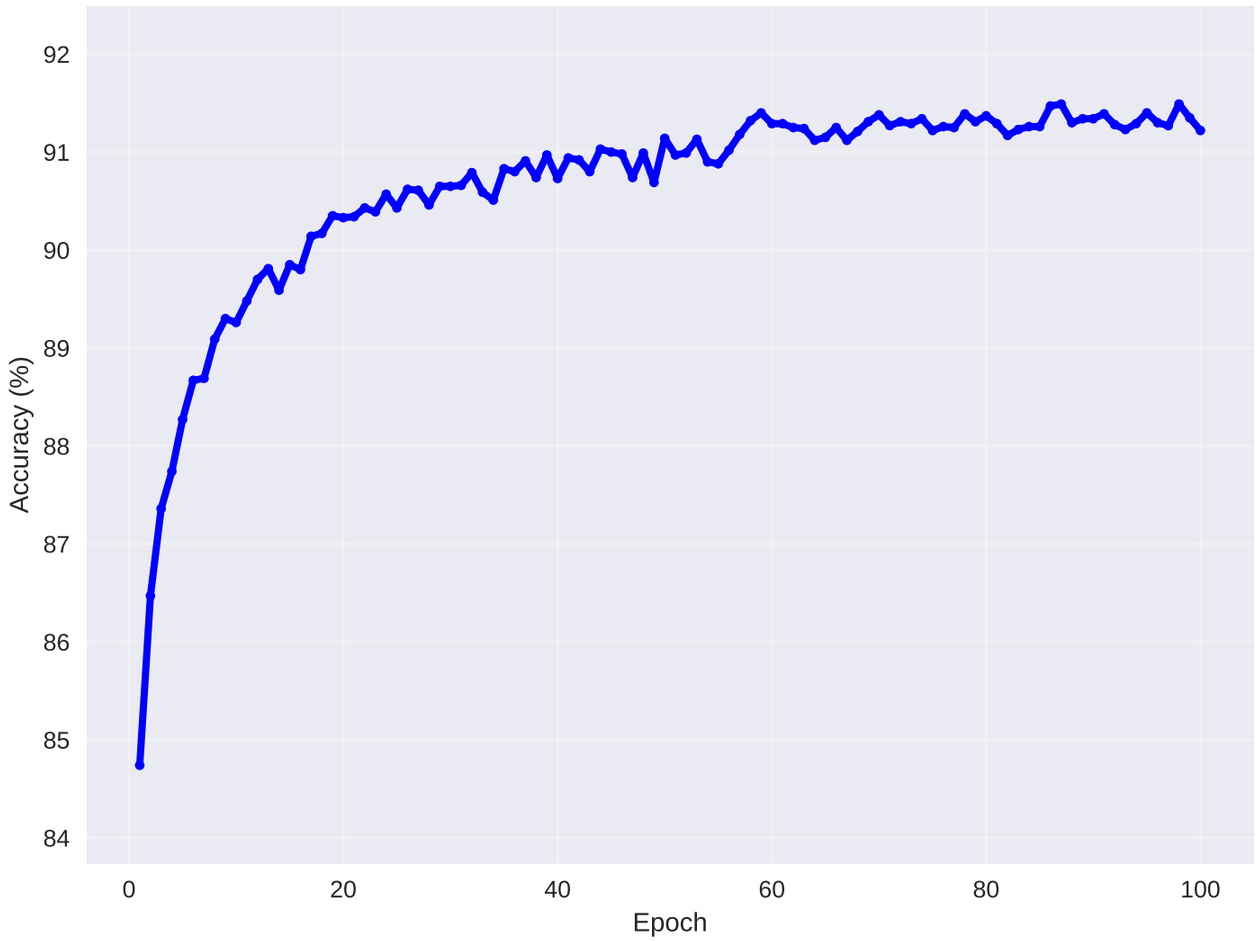
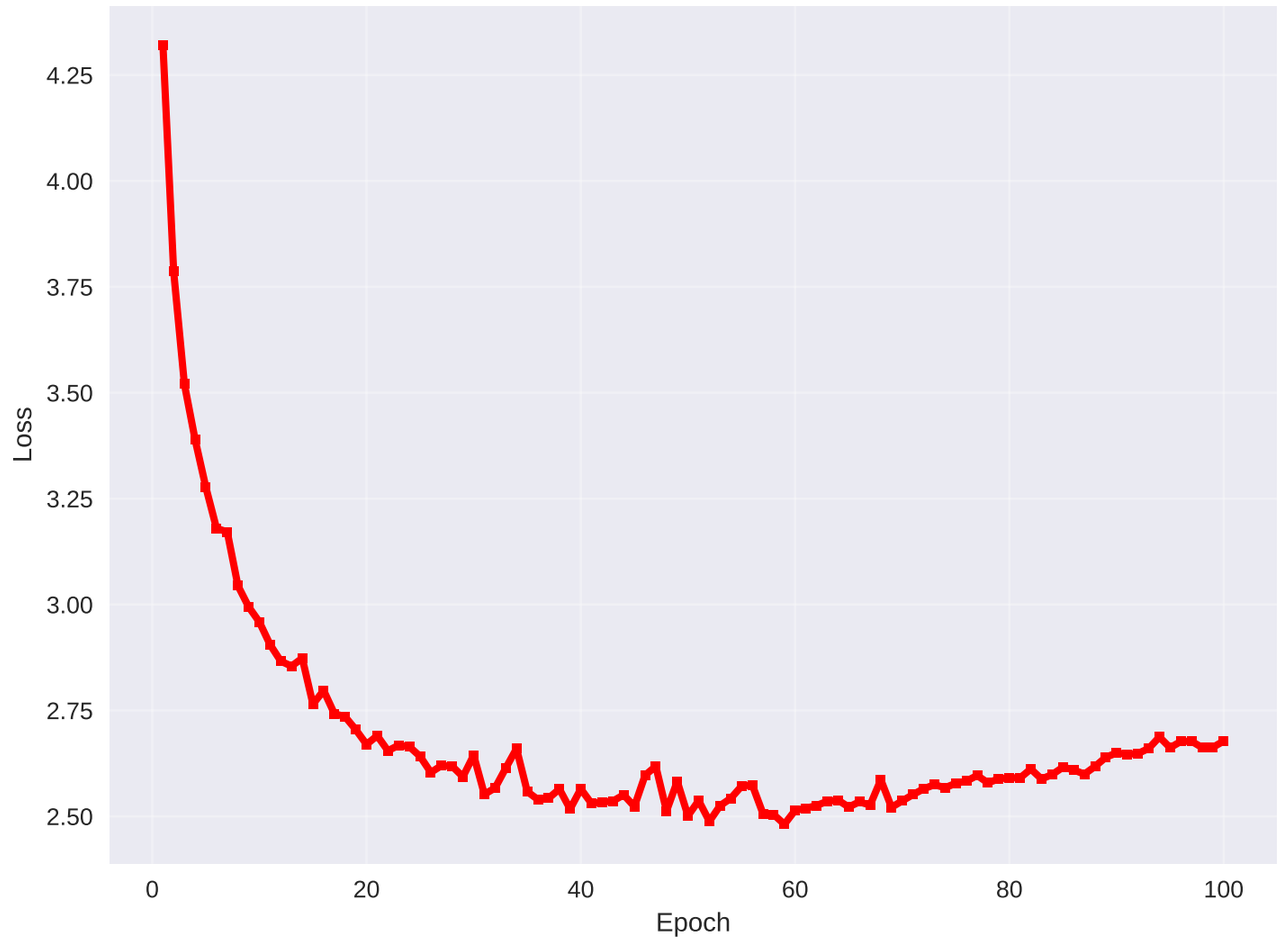
Federated Learning - Main Metrics Overview

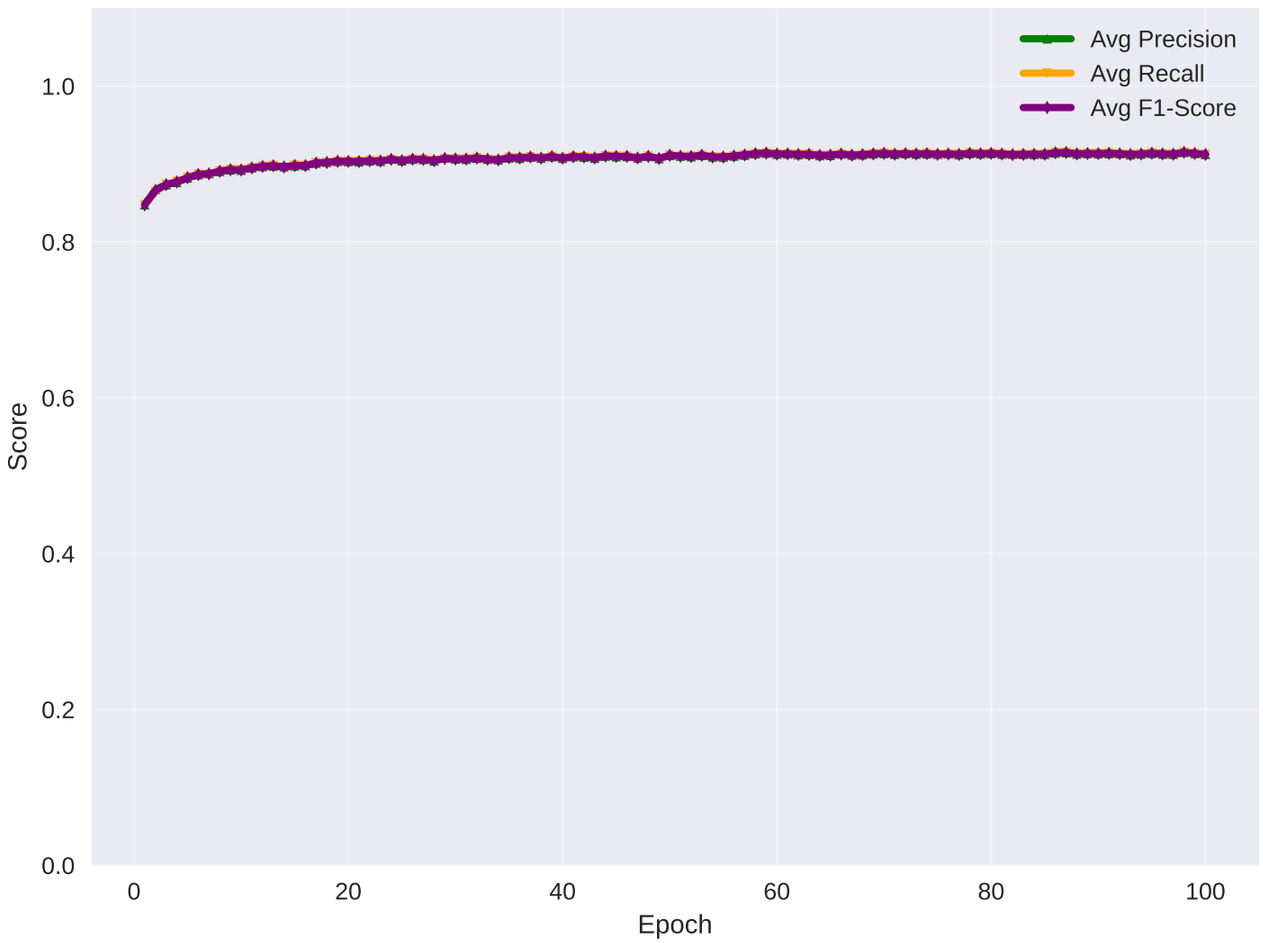
 Global Model Accuracy Evolution



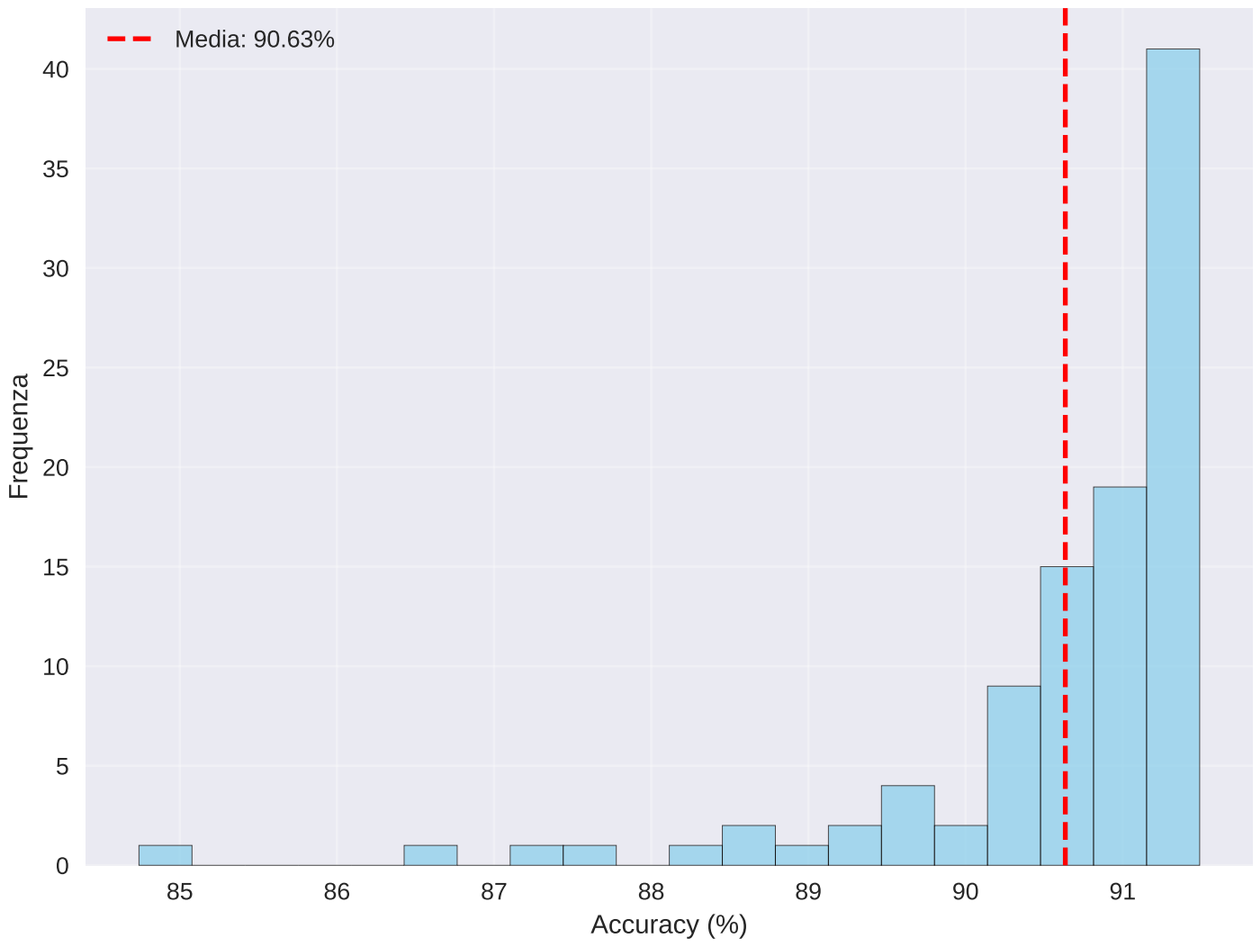
 Global Model Loss Evolution



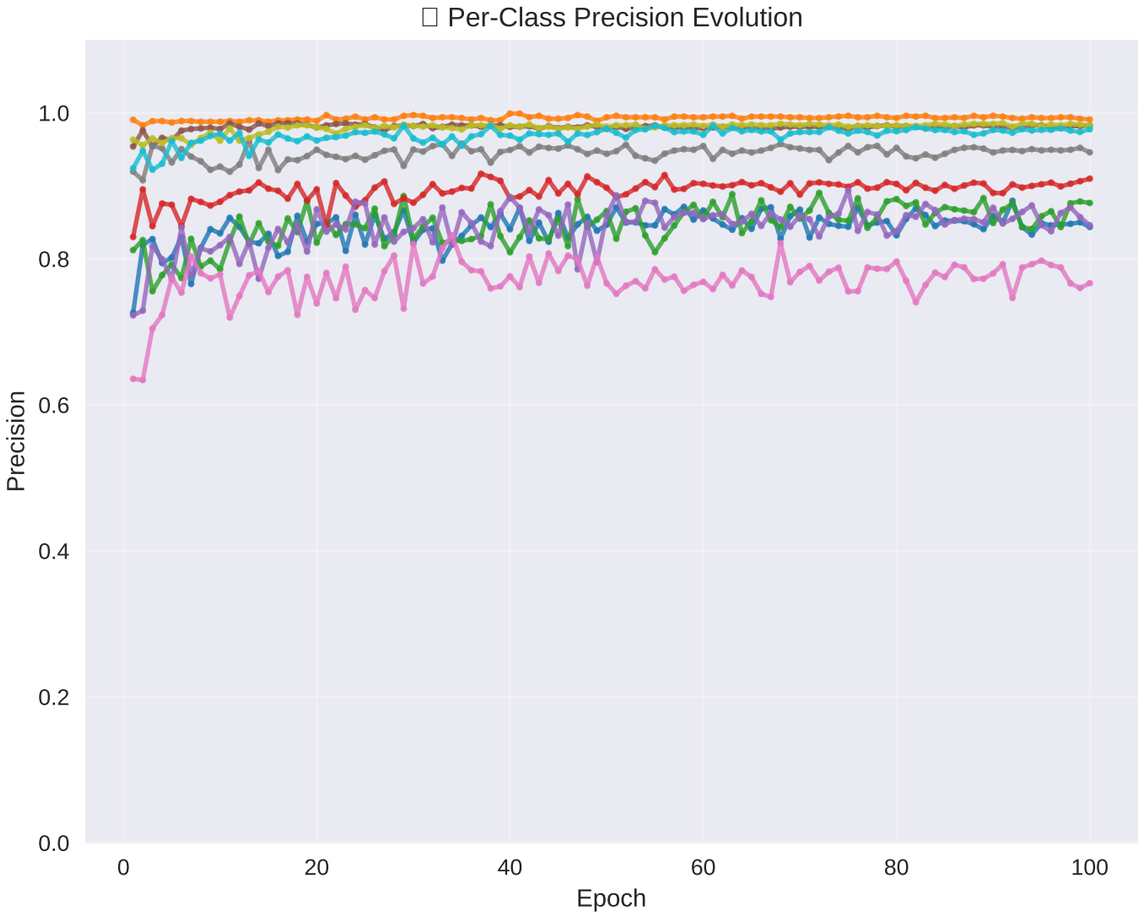
 Average Precision, Recall & F1-Score

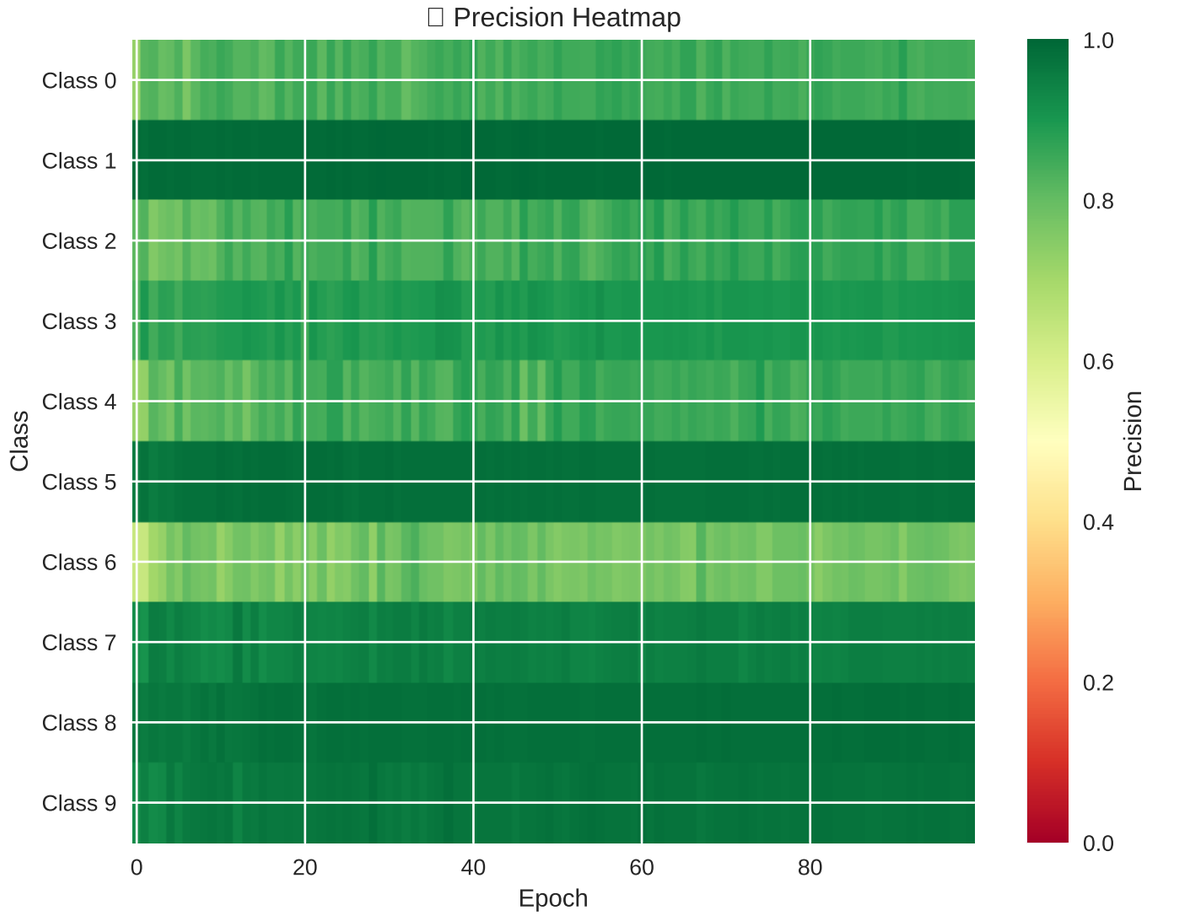


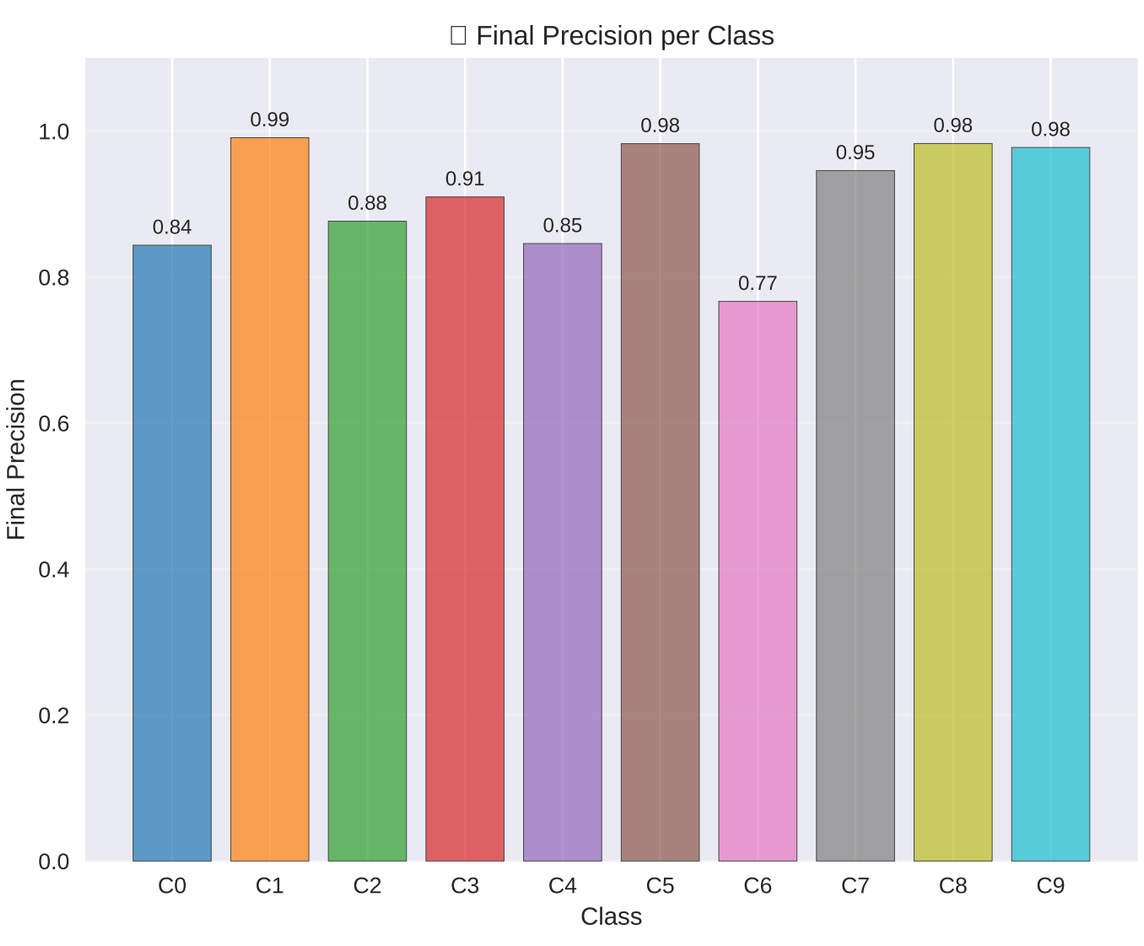
 Distribuzione Accuracy

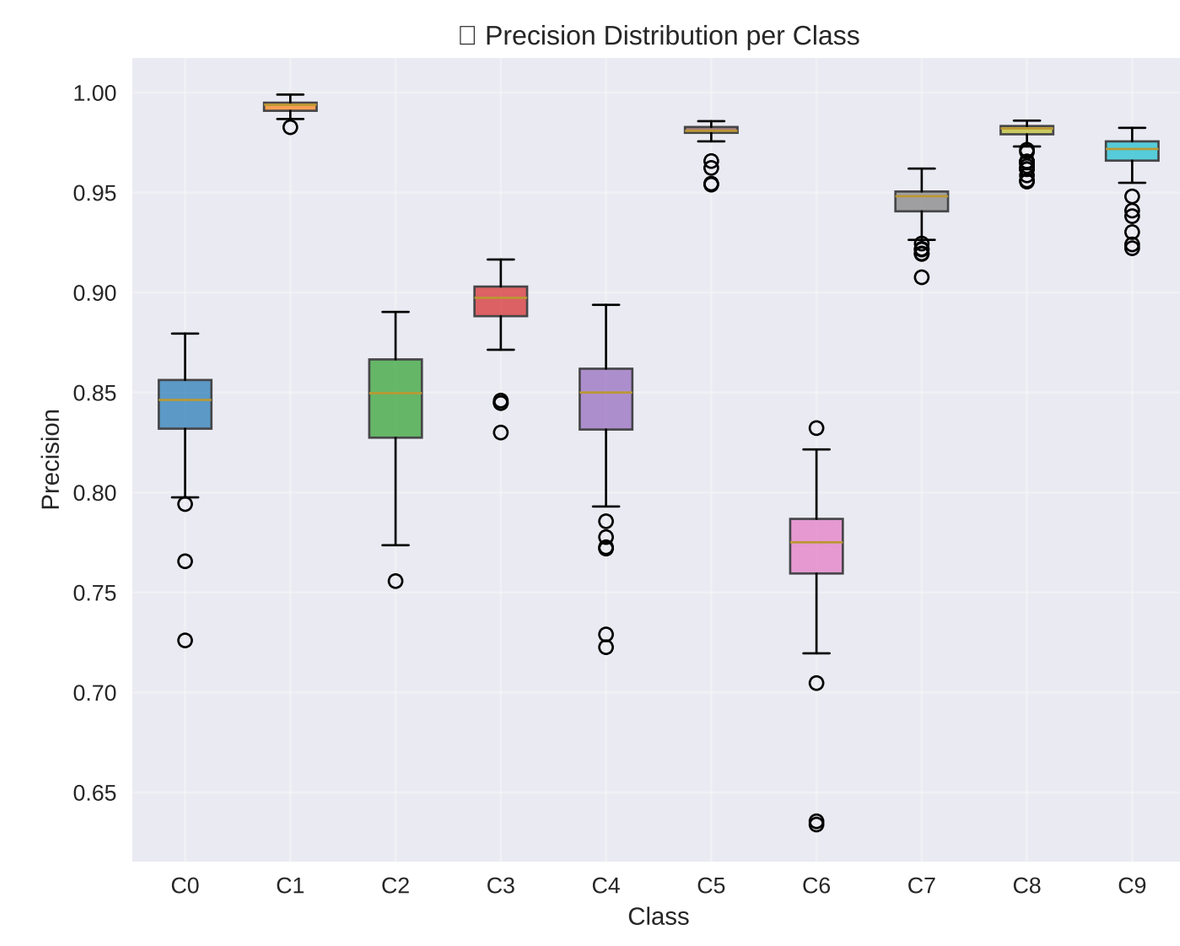


Per-Class Precision Analysis



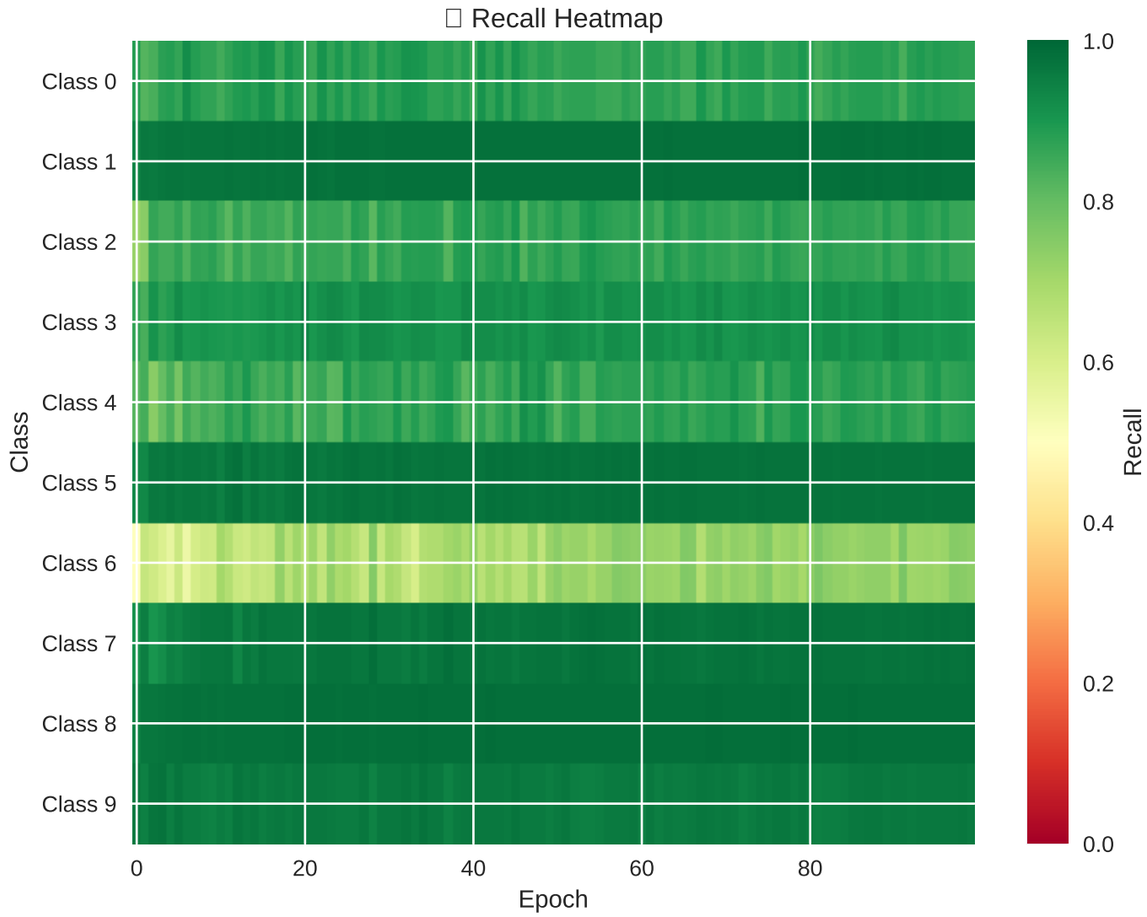


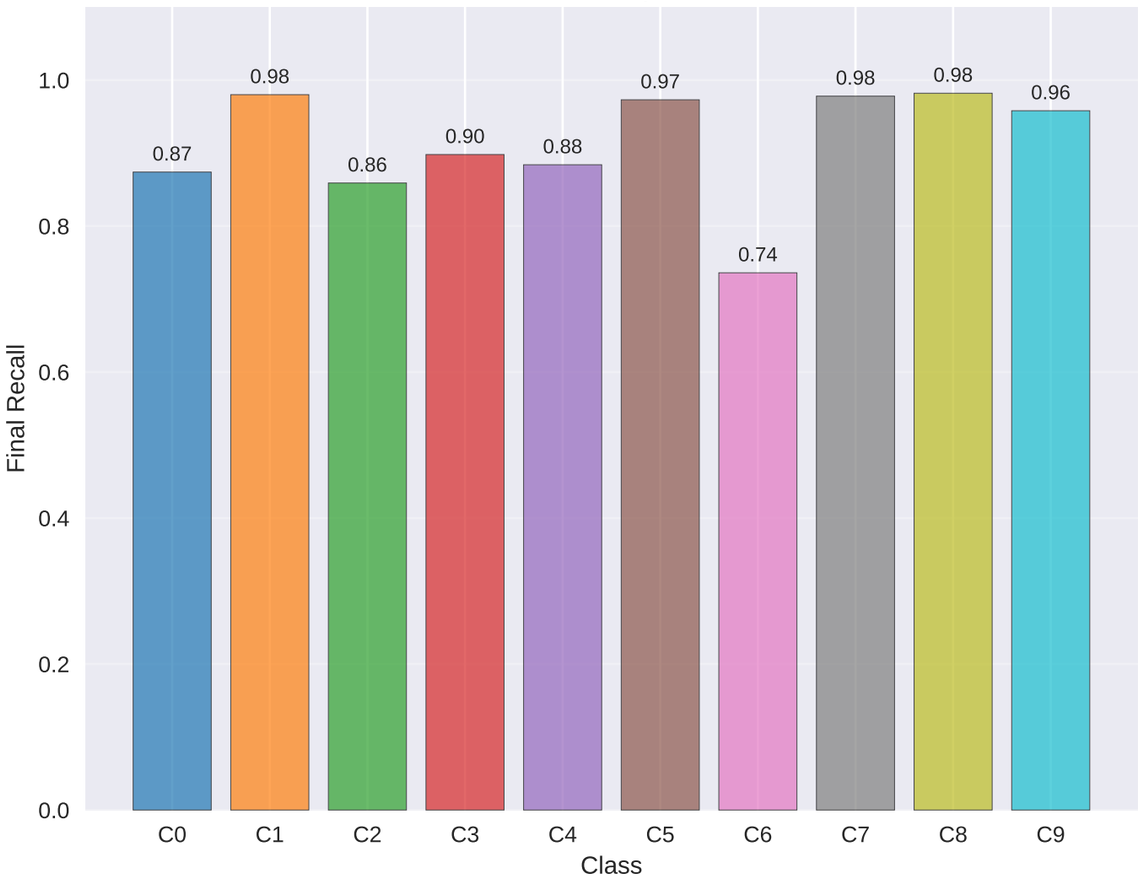


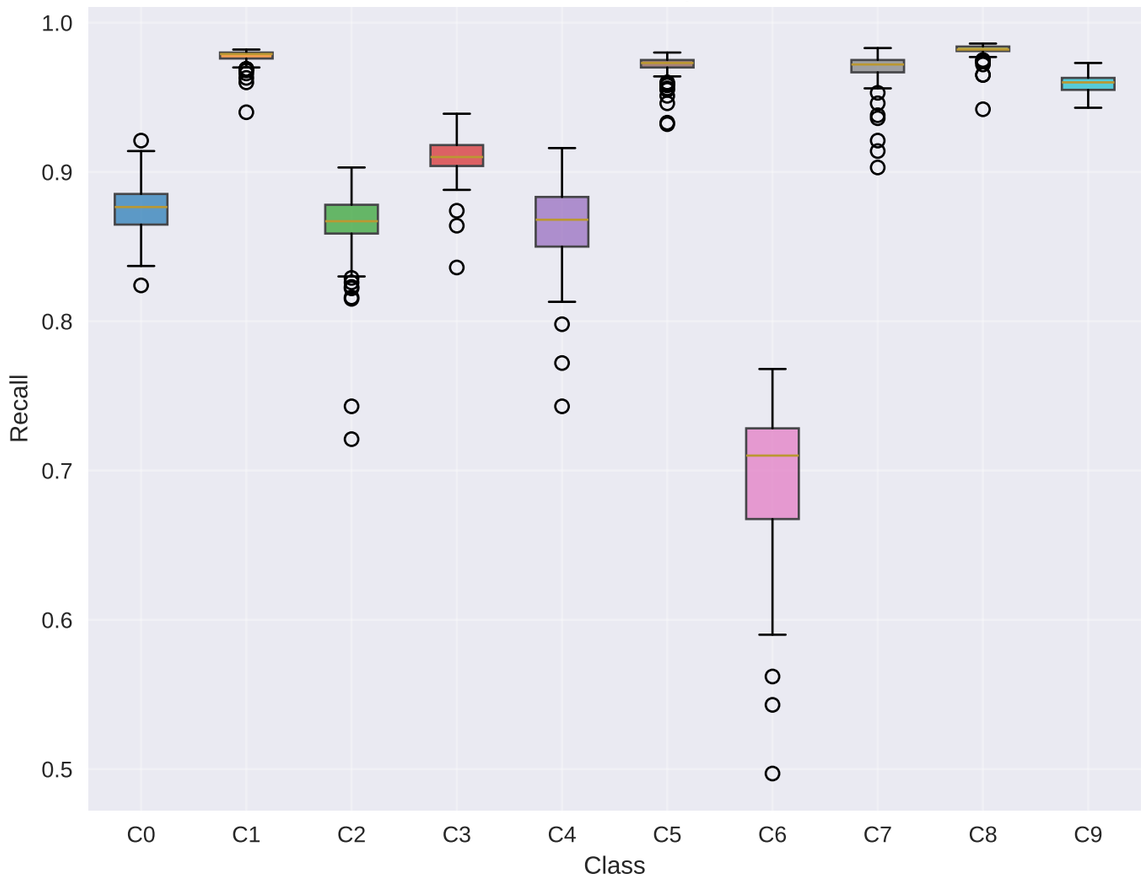


Per-Class Recall Analysis

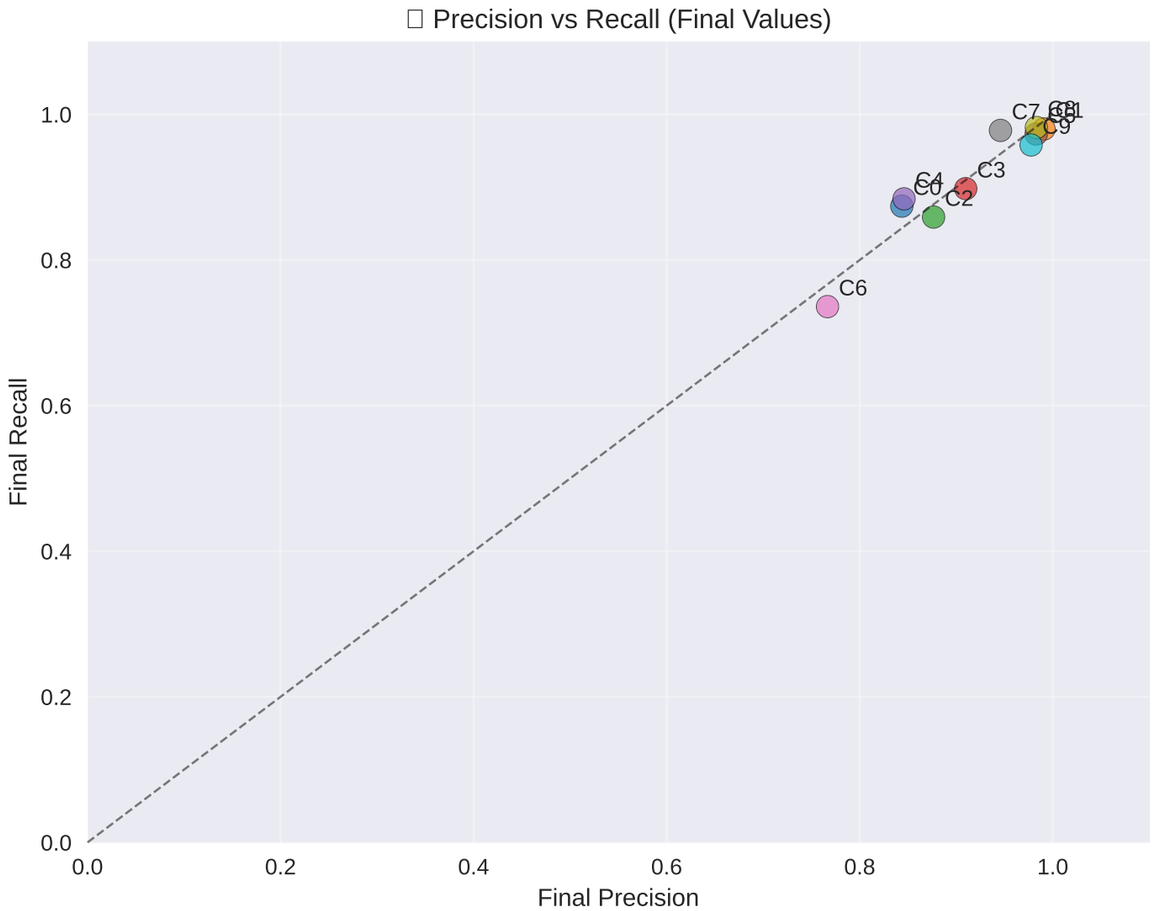




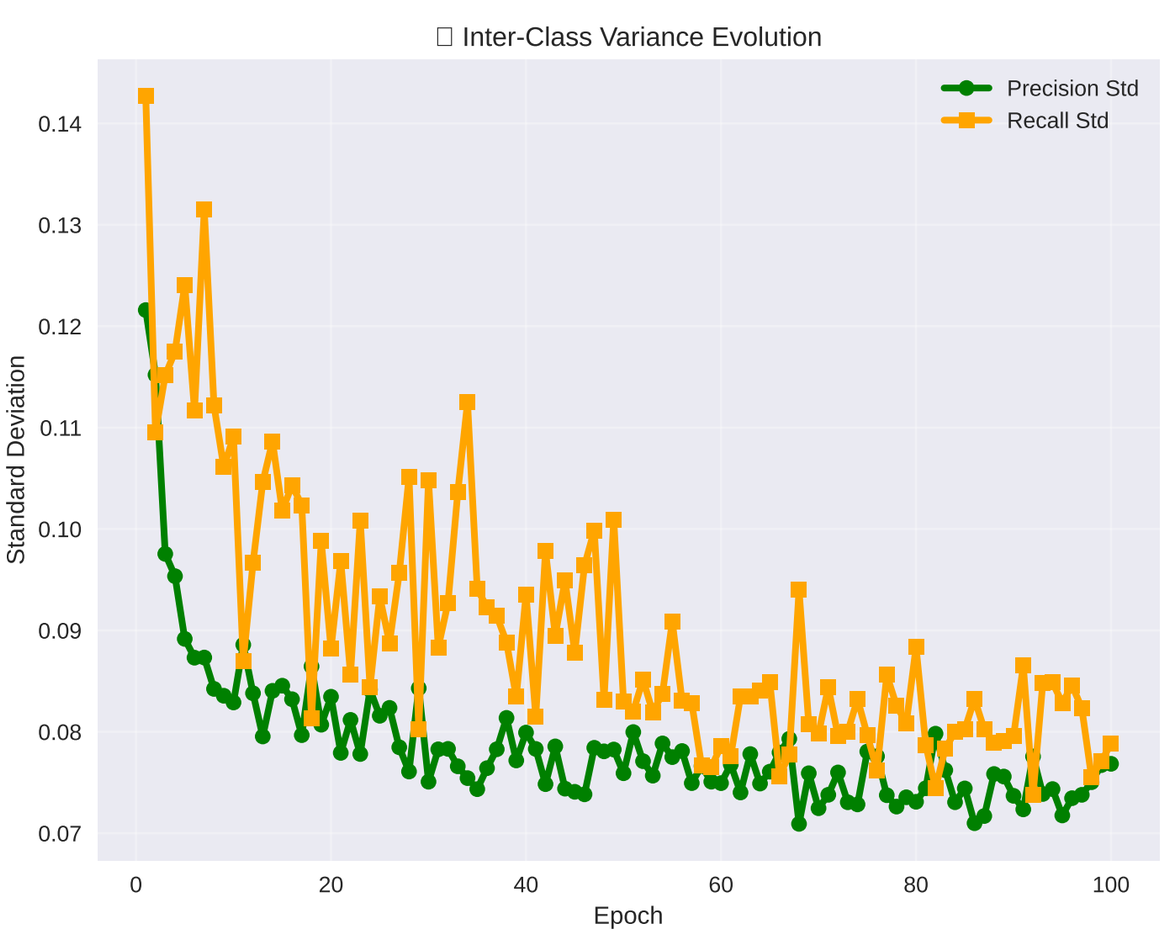


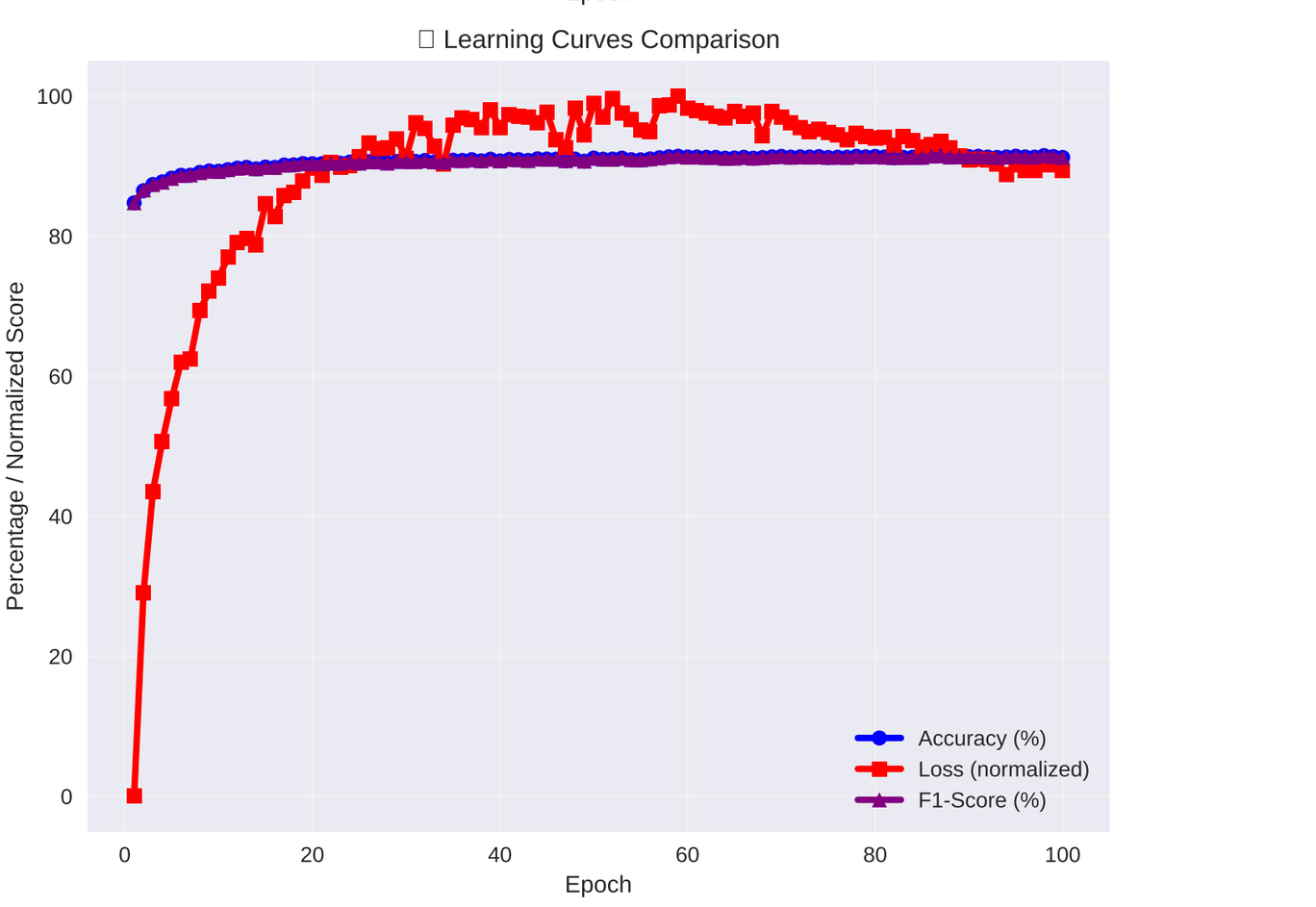


Comparative Analysis









Performance Summary Report

```
FEDERATED LEARNING - PERFORMANCE SUMMARY
========================================

FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-29 15:37:19
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.22%
• Loss: 2.6778
• Average Precision: 0.912
• Average Recall: 0.912
• Average F1-Score: 0.912

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.48%
• Riduzione Loss: +1.6431
• Stabilità Accuracy (CV): 0.0121

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.991)
• Peggiore Precision: Classe 6 (0.767)
• Migliore Recall: Classe 8 (0.982)
• Peggiore Recall: Classe 6 (0.736)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.844, R=0.874, F1=0.859
• Classe 1: P=0.991, R=0.980, F1=0.985
• Classe 2: P=0.877, R=0.859, F1=0.868
• Classe 3: P=0.910, R=0.898, F1=0.904
• Classe 4: P=0.846, R=0.884, F1=0.865
• Classe 5: P=0.983, R=0.973, F1=0.978
• Classe 6: P=0.767, R=0.736, F1=0.751
• Classe 7: P=0.946, R=0.978, F1=0.962
• Classe 8: P=0.983, R=0.982, F1=0.982
• Classe 9: P=0.978, R=0.958, F1=0.968

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
```